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Analysis in ideality factors at the single crystalline p+Si stacked on n−/n+Si interfaces for single-carrier control

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Abstract

In this work, a single-crystalline silicon nanomembrane (Si NM) was fabricated from a p+silicon-on-insulator wafer via an undercut process and transfer-printed onto n−/n+Si substrates to form vertically stacked pn heterostructures. Electrical analysis of the p+Si/n−/n+Si stacked structure configuration confirmed rectifying behavior with a turn-on at around 0.3–0.4 V and ideality factor (η) in the range of 1.2–1.3, consistent with diffusion and Shockley–Read–Hall recombination. In addition, a non-ideal region with $\eta > 2$ was observed, likely associated with interface traps or localized defects. This behavior pronounced at higher forward bias where increased carrier injection and decreased depletion width, allowing pre-existing traps to play a dominant role in the recombination process and leading to non-ideal carrier transport at the stacked interface. The analysis in the ideality factor at the stacked interface using a reliable method for reproducible pn junctions offers a foundation for extending this platform to advanced single-carrier devices and integration with lattice-mismatched semiconductors.

1. Introduction

Heterojunctions formed between lattice-mismatched semiconductors inevitably induce interfacial defects, which are known to degrade the electrical performance of devices [1–3]. In particular, for power devices operating at several kilovolts, it is essential to design heterostructures with precisely controlled band alignment [4]. However, lattice-mismatch makes it challenging to form heterojunctions using conventional epitaxial growth, resulting in deterioration of electrical properties and limitations in the choice of applicable material systems [5]. Since epitaxial processes are generally performed at high temperatures, interfacial interdiffusion or compositional non-uniformity may occur depending on the material combination, which can degrade the abruptness of the interface and impose additional constraints on device performance optimization [6]. Consequently, conventional growth- based approaches have difficulty avoiding lattice mismatch and the range of applicable material combinations tends to be limited by lattice matching requirements. In addition, certain material combinations are prone to quality degradation during growth and post-growth thermal treatment, which restricts their application to large-area fabrication. Therefore, a new stacking technology is required that can circumvent lattice-mismatch issue while integrating diverse materials [7].

Nanomembrane (NM) transfer technology has been proposed as a promising alternative to overcome these limitations [8, 9]. Since it does not require prolonged high temperature growth processes, interfacial degradation caused by thermal diffusion can be minimized and various combinations of materials and doping structures can be implemented with relatively high flexibility. These characteristics represent a key advantage over conventional epitaxial growth methods in terms of structural design flexibility and

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scalability. Moreover, NM transfer enables the fabrication of heterostructures with reduced defect densities by suppressing mismatch-induced dislocations and strain, offering advantages over conventional epitaxial growth [10–14]. In particular, single-crystalline Si NM has merits of low defect density and high compatibility with conventional CMOS process, allowing realization of stacked junctions and subsequent process integration on various substrates [15]. According to a recent report, silicon NMs, based on their ultrathin structure, can simultaneously provide excellent electrical properties and mechanical flexibility, and are emerging as a key platform for next-generation 'More-than-Moore' applications [16]. In addition, it has been reported that an amorphous interfacial region formed during Si/GaN stacking induces significant upward band bending, underscoring the critical role of interfacial bonding in determining band alignment [17].

In this study, these advantages were utilized to fabricate p+Si/n−/n+Si stacked structures, thereby implementing the most fundamental pn junction [18–20]. The p+Si/n–Si junction serves as a simple yet representative structure for evaluating the mechanism of charge carrier injection at the stacked interface, demonstrating the variation of ideality factors and associated recombination fundamentals. The single-carrier's behavior at the stacked interface was analyzed based on diffusion, Shockley–Read–Hall (SRH) recombination, and non-ideal transport. The p+Si/n−/n+Si structure can be expanded into fundamental architectures for various bipolar devices including power diodes and transistors, which verifies the applicability of proposed stacking approach. The results associated with the single-carrier's transport are readily extendable to actual integrated circuit environments.

2. Methods

The single-crystalline Si NM was fabricated from a p+silicon-on-insulator (SOI) wafer through an undercut process. The p+SOI wafer with a boron doping concentration of 1×10^{18} cm^{-3} was patterned by photolithography and inductively coupled plasma reactive ion etching (ICP-RIE) to form etching holes with dimensions of 5×5 μm^2 for Si NM formation, as shown in figure 1(a). The undercut process was carried out using HF: Acetone: DI water = 1:1:5 solution to remove the buried SiO_2 layer of the p+SOI wafer (figure 1(b)), thereby releasing the Si NM from the parent substrate, as shown in figure 1(c). The optical microscope image of the undercut p+SOI wafer is presented in figure 1(d). Using a polydimethylsiloxane (PDMS) stamp, the released Si NM was mechanically released from the p+SOI wafer and transfer-printed onto n−/n+Si substrates (figure 1(e)), forming stacked structures of p+Si/n−/n+Si, as shown in figure 1(f). Subsequently, rapid thermal annealing was performed at 600 °C for 3 min to complete the interfacial bonding. The optical microscope images of the completed stacked structures are shown in figure 1(g).

The thickness of the transferred Si NM was measured using a Dektak surface profiler. It was measured to be 250 nm for the p+Si/n−/n+Si stacked structure, confirming that the separation of the upper Si layer during the undercut process was carried out uniformly. To adjust the thickness of the Si NM in the completed structure and to deposit the electrodes, an etching process was performed using ICP-RIE. A mixed gas of SF_6 and Ar was used with the gas flow rates of 20 sccm. The etching conditions were set as follows: He pressure of 10 Torr, chamber pressure of 4 mTorr, source power of 300 W, bias power of 100 W, and an etching time of 15 s.

For the evaluation of electrical characteristics of the p+Si/n−/n+Si stacks, Ni/Au was deposited on the p-type top surface and Ti/Au was deposited on the n-type bottom surface of each structure to complete the electrode formation. Through this process, a pn junction stacked diode structure was realized. In the case of the p+Si/n−/n+Si stacked structure, a schematic illustration is shown in figure 2(a), where the vertically stacked diode consists of a p+Si NM (250 nm, 1×10^{18} cm^{-3}), an intermediate n–Si layer (500 nm), and a bottom n+Si layer (800 nm, 1×10^{18} cm^{-3}). The optical microscope image of the fabricated circular diode in figure 2(b) shows the central anode (Ni/Au, 40 μm in diameter) and the surrounding cathode (Ti/Au), confirming the successful formation of the pn diode structure.

3. Results and discussion

3.1. Rectifying characteristics

The electrical characteristics of the p+Si/n−/n+Si stacked structure based on transferred single-crystalline Si NM were measured using a semiconductor parameter analyzer. Figure 3 presents the *I–V* curves obtained from three independently fabricated samples procesamples fabricated through identical process. All samples exhibited typical rectifying behaviors, characterized by a sharp increase in forward current at around 0.3–0.4 V and effectively suppressed current under reverse bias, indicating that the p+Si/n−/n+Si stacked structure faithfully implements diode characteristics. Furthermore, the *I–V*

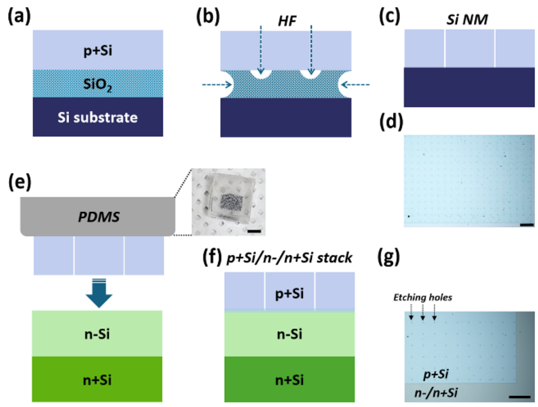

Figure 1. Process steps for the fabrication of single-crystalline Si nanomembrane (Si NM) and junction stacking with silicon substrates. (a) Preparation of a p+silicon-on-insulator (SOI) wafer with boron doping concentration of 1×10^{18} cm^{-3}, (b) undercut of p+SOI in HF solution, (c) undercut p+SOI wafer, (d) optical microscope image of released p+Si NM (scale bar = 100 μm), (e) transfer-printing of p+Si NM onto n−/n+Si substrates using a polydimethylsiloxane (PDMS) stamp. Inset shows a optical image of pick p+Si NM up using PDMS stamp (scale bar = 5 mm). (f) Structure of p+Si/n−/n+Si stack, followed by rapid thermal annealing at 600 °C for interfacial bonding, (g) and optical microscope image of the completed p+Si/n−/n+Si stack (scale bar = 100 μm).

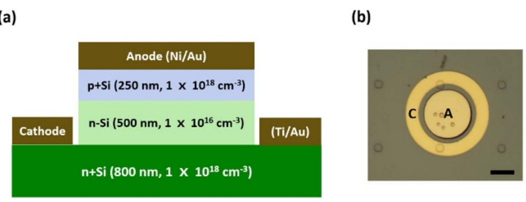

Figure 2. Device structure and electrode formation of p+Si/n−/n+Si stacked structure. (a) Schematic illustration of the vertically stacked diode consisting of a p+Si NM (250 nm, 1×10^{18} cm^{-3}), an intermediate n−Si layer (500 nm, 1×10^{16} cm^{-3}) and a bottom n+Si layer (800 nm, 1×10^{18} cm^{-3}) with Ni/Au electrodes deposited on the p-type top surface and Ti/Au electrodes deposited on the n-type bottom surface to form electrical contacts, (b) optical microscope image of the fabricated circular device with a central anode (A, Ni/Au, 40 μm in diameter) and surrounding cathode (C, Ti/Au), demonstrating realization of the pn diode structure based on the Si NM stacking process (scale bar = 20 μm).

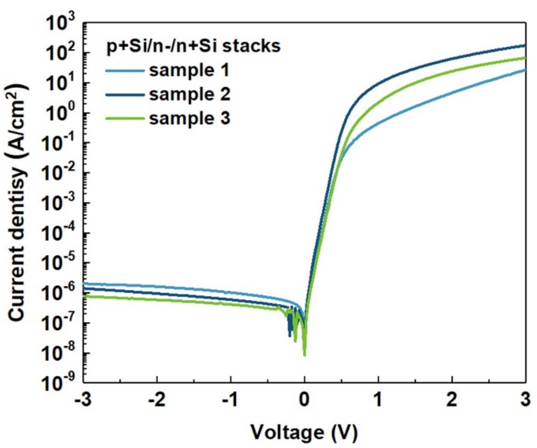

Figure 3. Electrical characteristics of the transferred single-crystalline Si NM-based stacked diodes with a p+Si/n−/n+Si structures configuration.

curves of the three samples showed highly similar shapes, confirming that identical rectifying behaviors can be reliably reproduced even after repeated processing. Among them, sample 2 demonstrated the highest forward current and the lowest reverse current, exhibiting relatively superior current characteristics. Such differences may be attributed to variations in interface quality or localized defect distributions, which will be further discussed in connection with the subsequent analysis of the ideality factor (η). Moreover, the consistent rectifying performance observed across the samples supports the reliability of the transfer and stacking process, suggesting that this structure provides a suitable platform for implementing stable diode operation in which bipolar carriers transfer across the stacked interface.

3.2. Ideality factor (η)

To quantitatively analyze the properties of the stacked junction, the ideality factor was extracted from the forward *I–V* characteristics to evaluate the charge carrier's behaviors. The ideality factor is an indicator of how well the carrier flows following the ideal diode equation. The ideal diode equation assumes that all recombination occurs either through band-to-band or trap-assisted recombination in the bulk region, in which case the ideality factor is 1,

$$I = I_0\left(\mathrm{e}^{\frac{qV}{nkT}} - 1\right). \tag{1}$$

However, in real devices, depletion region recombination, high-level injection, and interface trap-assisted recombination lead to deviations of the ideality factor. When the diffusion current is dominant, the ideality factor approaches 1, whereas when SRH recombination in the depletion region becomes dominant, the ideality factor approaches 2 [21, 22]. This can be attributed to the fact that pre-existing traps at the interface become more actively involved in the current transport as the applied voltage increases. As the forward bias increases, the depletion region narrows and the injected electron and hole densities rise. Consequently, the probability of recombination through traps distributed in the depletion region and at the interface increases. As a result, unlike the low voltage regime dominated by diffusion current, in the high voltage regime SRH recombination within the depletion region becomes the dominant mechanism, causing the ideality factor to approach 2 [23–25].

In figure 4, the ideality factor was extracted as a function of voltage for the p+Si/n−/n+Si stacked structure, and the current transport mechanisms were analyzed according to the voltage regions. In region A (∼0.1 V), the ideality factor is generally close to 1. This indicates that diffusion current is the major transport mechanism. SRH recombination may occur concurrently due to the influence of interface traps in the depletion layer, resulting in a slightly higher η (1.1–1.2) compared to the ideal value of 1. In region B (0.1–0.4 V), the ideality factor of samples 2 and 3 are distributed overall in the range of 1.2–1.3. This implies that diffusion current dominates the current transport mechanism. The current is mainly determined by minority carrier diffusion and the contribution of SRH recombination is limited. The p+Si/n−/n+Si structure exhibits ideal pn junction behavior and the stacked junction interface can be interpreted as having a low interface trap density. The sample 1 shows a slightly larger η value in the same voltage region, indicating that SRH recombination occurs due to interface non-uniformity. In region C (>0.4 V), the ideality factor of sample 1 significantly increases beyond 2. This behavior cannot be explained solely by the SRH recombination and is instead interpreted as a phenomenon induced by local interface defects or surface damage. Moreover, the increase in carrier injection density and the narrowing of the depletion region with rising voltage enhance the contribution of pre-existing interface traps and extended defects, leading to dominant multi-level recombination and the manifestation of non-ideal transport characteristics [26]. On the other hand, samples 2 and 3 maintained lower η values across the entire voltage range compared to sample 1, indicating superior interface quality.

$$R_{\mathrm{SRH}} = \frac{n_{\mathrm{f}}p_{\mathrm{f}}}{\frac{n_{\mathrm{f}}}{\sigma_p} + \frac{p_{\mathrm{f}}}{\sigma_n}} v_{\mathrm{th}} N_{\mathrm{t}} = \left(\frac{1}{\sigma_p p_{\mathrm{f}}} + \frac{1}{\sigma_p n_{\mathrm{f}}}\right)^{-1} v_{\mathrm{th}} N_{\mathrm{t}} \tag{2}$$

The experimentally observed voltage dependence of the ideality factor can be interpreted based on equations (1)–(3). Region A can be explained by a diffusion–dominated current mechanism. In Region B, the behavior is consistent with minority–carrier–controlled SRH recombination, and can be explained by the conventional SRH model of equation (2) [27]. In contrast, the observation of η exceeds 2 in Region C suggests that recombination may no longer be entirely limited by minority carriers. In the presence of traps, field enhanced recombination mechanisms including trap–assisted tunneling (TAT) may become active under finite electric fields. Under this conditions, TAT can enhance the trap-related recombination component and recombination models incorporating this effect have been reported to reproduce ideality factors significantly exceeding 2 [28],

$$R_{\mathrm{trap}} = \frac{pn - n_{ie}^2}{\frac{\tau_p}{1+\Gamma_p}\left[n + n_{ie}\exp\left(\bar{E}/kT\right)\right] + \frac{\tau_n}{1+\Gamma_n}\left[p + n_{ie}\exp\left(-\bar{E}/kT\right)\right]}. \tag{3}$$

Here, Γ_p and Γ_n denote electric field dependent tunneling enhancement factors. While equation (3) reduces to the conventional SRH expression in the weak field regime, the trap related recombination

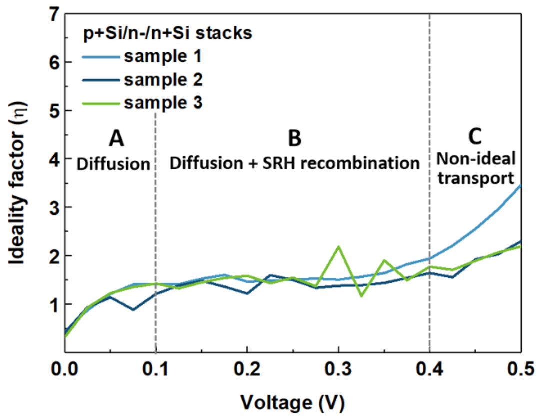

Figure 4. Ideality factor–voltage curves extracted from the p+Si/n−/n+Si stacked structure. In region A (V < 0.1 V), the ideality factor remains at approximately 1.0–1.2, indicating diffusion-dominated transport with minor influence from interface traps. In region B (0.1–0.4 V), the ideality factor slightly increases ($\eta \approx 1.2$–1.5), suggesting that, in addition to diffusion, weak Shockley–Read–Hall (SRH) recombination contributes due to interface non-uniformity. In region C ($V > 0.4$ V), the ideality factor exceeds 2, reflecting non-ideal transport behavior caused by interface traps and surface-related defects. This can be interpreted as the dominant action of multi-level recombination through pre-existing interface traps and extended defects, facilitated by increased carrier injection and narrowing of the depletion region with rising voltage,

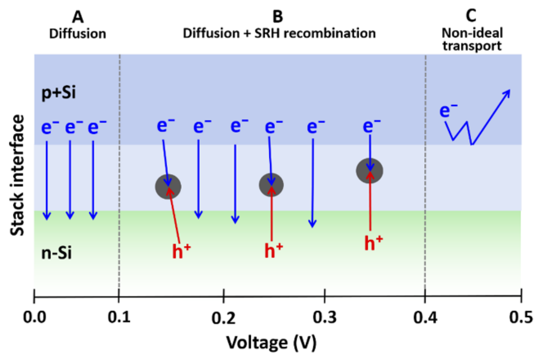

Figure 5. Schematic illustration of carrier transport mechanisms at the p+Si/n−/n+Si stacked junction under different voltage regions. In region A, electrons diffuse directly across the junction, resulting in an ideal current flow. In region B, SRH recombination is involved, where electrons (e^-) are captured by interface traps and holes (h^+) move to the same sites to recombine. In region C, interface traps or surface defects hinder direct transport, leading to detoured or leakage current paths that cause non-ideal current flow.

term can be substantially enhanced when Γ_p and Γ_n become significant under finite electric fields, which offers a theoretical framework for interpreting ideality factors greater than 2.

Figure 5 schematically illustrates the carrier transport mechanisms corresponding to the voltage-dependent regions observed in the ideality factor curve. In region A, electrons primarily followed a direct diffusion path across the junction, which corresponds to ideal diode operation with the ideality factor approaching unity. In region B, an additional process was observed in which electrons and holes recombined through interface traps via the SRH recombination mechanism. This is indicated by trap symbols (●) and recombination sites in the schematic, and the ideality factor increased to approximately 1.2–1.5 in this region [29, 30]. In region C, localized interface defects or surface damage caused current to deviate from a straightforward path and follow detoured conduction pathways. These non-ideal transport routes were represented by curved arrows in the schematic, and the ideality factor exceeded 2 in this region. The $\eta > 2$ region observed in this study can generally be interpreted as non-ideal behavior originating from interface traps or surface defects. Since the presence and density of interface traps were not directly quantified, this interpretation is based on indirect evidence. Nevertheless, previous studies have reported that interface traps are not only detrimental to device performance but can also be associated with novel charge transport phenomena. For example, single-electron trapping has been observed in Si lateral pn junctions and single-electron charging and Coulomb blockade effects have been demonstrated in quantum dot–embedded Schottky diodes [31, 32]. These phenomena represent fundamental charge-control mechanism relevant to quantum information technologies. Thus, under specific conditions, interface traps may act as localized charge storage nodes or serve as a basis for single-carrier control [33].

Although the $\eta > 2$ region observed in this study cannot be regarded as direct evidence of carrier control, it can be understood as an experimental observation indicating that interface traps can significantly alter current transport behavior. In addition, the transferred single-crystalline Si NM stacked structure consistently exhibited stable rectifying characteristics even after repeated processing, confirming process reproducibility and structural reliability. In particular, a recent XPS-based heterojunction study reported experimentally observed large ideality factors, suggesting that interfacial band offsets and defect states can significantly influence current transport behavior [34]. The non-ideal behavior in the $\eta > 2$ region suggests the potential for extending this structure toward single-carrier control and quantum transport research through further verification, such as low-temperature *I–V* measurements, time-resolved current analysis, and capacitance-based trap characterization.

4. Conclusions

In conclusion, single crystalline p+Si/n−/n+Si stacked structures were successfully fabricated through the NM transfer process, and electrical measurements confirmed that stable rectifying characteristics were reproducibly observed in all three samples. The ideality factor was distributed in the range of 1.2–1.3 as typical pn junction behavior, while a non-ideal region with $\eta > 2$ was observed at certain voltage ranges, suggesting the influence of interface traps or localized defects. These results indicate that the proposed process is effective for forming reliable stacked heterojunctions for single-carrier's control, with the analysis of recombination mechanism. The NM transfer-based stacking approach suggests the potential for future extension to carrier transport studies for next-generation quantum computing applications.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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